

Scrambling in Japanese: Hidden Similarities to Topicalization

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Abstract

The purpose of this paper is to establish two novel claims on the nature of scrambling in Japanese. First, as opposed to the general consensus that has been reached since Saito (1992) and Tada (1993), we show that not only clause-internal (CI) scrambling, but also long-distance (LD) scrambling can create new binding/scope relations, so the latter need not undergo radical reconstruction, either. Second, we argue that, though CI- and LD-scrambling can both have semantic effects, they are still not uniform in nature and should be derived by distinct grammatical processes, contra Tada (1990), Kitahara (2002), and Saito (2003). In particular, we show that, for LD-scrambling to apply, it must obey the same contrast condition as topicalization in English does, but CI-scrambling need not. To derive this disparity, we adopt our recent idea in Shimamura and Tanaka (to appear), under which LD-scrambling is always a kind of topicalization, but CI-scrambling can be derived, not only by topicalization, but also by base-generation. We lend further support to this dual identity of CI-scrambling by examining the behavior of VP-internal scrambling, based on some of Nemoto's (1993) findings.

1. Introduction

Since Hoji (1985) and Saito (1989), it has been assumed that scrambling in Japanese may be semantically undone via a procedure called *reconstruction*. This means that scrambling, whether clause-internal (CI) or long-distance (LD), can be an operation that moves an element to a higher position, but allows the element to get interpreted in its base position. One diagnosis to exemplify reconstruction is variable binding. In (1B) and (2B), for example, the object containing the pronoun *soko*₁ ‘there/it’ undergoes CI- and LD-scrambling.

(1) CI-scrambling

- A: Zyaa, kaku-kigyoo₁-ga [*soko*₁-no-KAITYOO]₂-o uttaeta-no?
then every-company-Nom there-of-chairperson-Acc sued-Q
‘Then, did every company₁ sue [its₁ chairperson]?’
- B: Iya, [*soko*₁-no-SHATYOO]₂-o kaku-kigyoo₁-ga *t*₂ uttaeta-ndayo.
no there-of-CEO-Acc every-company-Nom sued-Emp(hasis)
‘No, [its₁ CEO]₂, every company₁ sued *t*₂.’

(2) LD-scrambling

- A: Zyaa, kaku-kigyoo₁-ga [[*soko*₁-no- KAITYOO]₂-o uttaeta]-to happyoo-sita-no?
then every-company-Nom there-of-chairperson-Acc sued-C announcement-did-Q
‘Then, did every company₁ announce that it sued [its₁ chairperson]?’

- B: Iya, [soko₁-no-SHATYOO]₂-o kaku-kigyoo₁-ga [t₂ uttaeta]-to happyoo-sita-ndayo.
 no there-of-CEO-Acc every-company-Nom sued-C announcement-did-Emp
 ‘No, [its₁ CEO]₂, every company₁ announced that it sued t₂.’

In both cases, the pronoun *soko*₁ in the scrambled object can be construed as a variable bound by the quantifier ‘every company.’ Given that bound variables must be c-commanded by their quantifiers at the semantic level, the availability of such readings indicates that CI- and LD-scrambling both allow the scrambled object to be interpreted in its base position. They are therefore uniform in exhibiting the reconstruction property.

In contrast to this similarity, Saito (1992) and Tada (1993) have argued that the two types of scrambling differ in one respect. That is, LD-scrambling must be “A’-movement,” while CI-scrambling need not, and LD-scrambled elements fail to bind into or take scope over a matrix element. Probably the most cited diagnosis to show the difference is binding of the reciprocal pronoun *otagai*, which is often translated as ‘each other.’ (3) and (4) are borrowed from Saito (1992: 74-76) along with his acceptability judgment.

(3) CI-scrambling

- a. ?*Otagai₁-no-sensei-ga karera₁-o hihan-sita.
 e.o-of-teacher-Nom they-Acc criticism-did
 ‘Each other₁’s teachers criticized them₁.’
 b. ? Karera₁-o otagai₁-no-sensei-ga t₁ hihan-sita.
 they-Acc e.o-of-teacher-Nom criticism-did
 ‘Them₁, each other₁’s teachers criticized t₁.’

(4) LD-scrambling

- a. * Otagai₁-no-sensei-ga [Hanako-ga karera₁-o hihan-sita]-to itta.
 e.o-of-teacher-Nom Hanako-Nom they-Acc criticism-did-C said
 ‘Them₁, each other₁’s teachers said that Hanako criticized t₁.’
 b. * Karera₁-o otagai₁-no-sensei-ga [Hanako-ga t₁ hihan-sita]-to itta.
 they-Acc e.o-of-teacher-Nom Hanako-Nom criticism-did-C said
 ‘Them₁, each other₁’s teachers said that Hanako criticized t₁.’

According to Saito, the scrambled object can bind *otagai* in (3b), but that is impossible in (4b), where LD-scrambling takes place.

Given the reported disparity, suppose that LD-scrambling always behaves as “A’-movement,” in the sense that it lacks the ability to yield a new binding/scope relation. Then, the question is why it does. A solution has been developed by Saito (2003), particularly based on Tada (1990) and Kitahara (2002). The leading idea is to reduce the “A’-property” to reconstruction, assuming that LD-scrambling obligatorily reconstructs, but not CI-scrambling. To deduce this distinction, Saito adopts the *Copy Theory of Movement* (Chomsky 1995) and proposes that copies of a moved phrase are derivationally interpreted and deleted. Specifically, he claims that, as a result of his system, the higher copy created by LD-scrambling has its semantic content deleted, so the syntax ignores it and sends the lower copy of the scrambled phrase for interpretation. Under Saito’s theory, therefore, CI- and LD-scrambling are unified as a single movement operation, and the “A’-property” of the latter falls out as a by-product of the obligatory deletion of the higher copy’s semantic content.

In this paper, however, we will critically reconsider the empirical basis of such a unified approach and instead argue for a “hybrid” approach to scrambling in Japanese. To this end, the discussion goes as follows.

In Sections 2 and 3, we first show that LD-scrambling, like CI-scrambling, need not be semantically undone, yet they are not uniform in nature and must be derived by different grammatical processes. We then adopt our recent proposal in Shimamura and Tanaka (to appear) and claim that, while LD-scrambling is a case of a movement operation which we identify with *topicalization* (cf. Vermeulen 2012, Abe 2022, Mihara 2023), CI-scrambling can be derived not only by topicalization, but also by *base-generation*. To support this proposal, Section 4 compares LD-scrambling and English topicalization, showing that they display the same range of semantic effects. Turning then to CI-scrambling, Section 5 verifies its dual identity by considering the behavior of VP-internal scrambling in the ditransitive construction (e.g., Nemoto 1993). Section 6 concludes.

2. Against Obligatory Syntactic Reconstruction

We begin by showing that the adoption of obligatory syntactic reconstruction in the analysis of LD-scrambling is not valid for two reasons. Our first concern is the acceptability status of (4b), which has been used to motivate the “A’-property” of LD-scrambling. Of course, we agree with Saito’s judgment, and in (4b) it is much harder to get the bound reading of *otagai* than in (3b). Still, Saito (1992: 111, n.8) himself notes that some speakers accept such examples as (4b), and Mihara (2023: 221ff.) also reports that adding some stylistic revisions to (4b) improves its status. (5) and (6) are our own examples, and in (6b), we find it much easier to construe the LD-scrambled object as the binder of *otagai* than in (4b).

(5) CI-scrambling

- a. * *Otagai*₁-no-sensei-ga TA-tosite [**Ken-to-Yui**]₁-o yatotta.
 e.o-of-teacher-Nom TA-as Ken-&-Yui-Acc hired
 ‘Each other₁’s teachers hired [Ken and Yui]₁ as TAs.’
- b. [**Ken-to-Yui**]₁-o *otagai*₁-no-sensei-ga TA-tosite *t*₁ yatotta.
 Ken-&-Yui-Acc e.o-of-teacher-Nom TA-as hired
 ‘[Ken and Yui]₁, each other₁’s teachers hired *t*₁ as TAs.’

(6) LD-scrambling

- a. * *Otagai*₁-no-sensei-ga [TA-tosite [**Ken-to-Yui**]₁-o yatotta]-to itta.
 e.o-of-teacher-Nom TA-as Ken-&-Yui-Acc hired-C said
 ‘Each other₁’s teachers said that they hired [Ken and Yui]₁ as TAs.’
- b. [**Ken-to-Yui**]₁-o *otagai*₁-no-sensei-ga [TA-tosite *t*₁ yatotta]-to itta.
 Ken-&-Yui-Acc e.o-of-teacher-Nom TA-as hired-C said
 ‘[Ken and Yui]₁, each other₁’s teachers said that they hired *t*₁ as TAs.’

The same point can also be made by other types of diagnoses. For example, (7) and (8) are cases of quantifier binding, and (8b) shows that the LD-scrambled object can bind the pronoun *soko*₁ in the matrix subject.

(7) CI-scrambling

- a. * *Soko*₁-no-syain-ga tuini **kaku-burakku-kigyoo**₁-o uttaeta.
 there-of-staff-Nom finally every-toxic-company-Acc sued
 ‘The staff there₁ finally sued every toxic company₁.’
- b. **Kaku-burakku-kigyoo**₁-o *soko*₁-no-syain-ga tuini *t*₁ uttaeta.
 every-toxic-company-Acc there-of-staff-Nom finally sued
 ‘Every toxic company₁, the staff there₁ finally sued *t*₁.’

(8) **LD-scrambling**

- a. * Soko₁-no-syain-ga tuini [kaku-burakku-kigyoo₁-o uttaeru]-to itta.
 there-of-staff-Nom finally every-toxic-company-Acc sue-C said
 ‘The staff there₁ finally said that they would sue Every toxic company₁.’
- b. **Kaku-burakku-kigyoo₁-o** soko₁-no-syain-ga tuini [t₁ uttaeru]-to itta.
 every-toxic-company-Acc there-of-staff-Nom finally sue-C said
 ‘Every toxic company₁, the staff there₁ finally said that they would sue t₁.’

Next, (9) and (10) involve scope interaction. In (9b) and (10b), the universal quantifier ‘every grad’ undergoes CI- and LD-scrambling, and it can take scope over the existential quantifier ‘one of the teachers.’ In other words, it is possible to assign both examples a surface scope reading (i.e., $\forall > \exists$).¹

(9) **CI-scrambling**

- a. Sensei-no-dareka-ga TA-tosite **kaku-insei-o** yatotta.
 teacher-of-some-Nom TA-as every-grad-Acc hired
 ‘One of the teachers hired every grad₁ as a TA.’ (* $\forall > \exists$)
- b. **Kaku-insei₁-o** sensei-no-dareka-ga TA-tosite t₁ yatotta.
 every-grad-Acc teacher-of-some-Nom TA-as hired
 ‘Every grad₁, one of the teachers hired t₁ as a TA.’ ($\forall > \exists$)

(10) **LD-scrambling**

- a. Sensei-no-dareka-ga [TA-tosite **kaku-insei₁-o** yatotta]-to itta.
 teacher-of-some-Nom TA-as every-grad-Acc hire-C said
 ‘One of the teachers said that s/he hired every grad₁ as a TA.’ (* $\forall > \exists$)
- b. **Kaku-insei₁-o** sensei-no-dareka-ga [TA-tosite t₁ yatotta]-to itta.
 every-grad-Acc teacher-of-some-Nom TA-as hire-C said
 ‘Every grad₁, one of the teachers said that s/he hired t₁ as a TA.’ ($\forall > \exists$)

The “A’-property” of LD-scrambling therefore is not as robust as reported in the literature. If our observations are correct, then the use of obligatory syntactic reconstruction is not tenable, since it fails to capture the fact that the LD-scrambled objects in (6b), (8b), and (10b) bind into or take scope over the matrix subjects.

At this point, it is worth clarifying what differences exist between “traditional” examples such as (4b) and ours, influencing the interpretation of LD-scrambling. The most crucial difference is that in our examples, the

¹ It is unclear whether (9b) and (10b) also have a reconstructed reading (i.e., $\exists > \forall$), since any situation that makes such a reading true also accepts the surface reading (i.e., $\forall > \exists$); that is, if the $\exists > \forall$ reading is true for one situation, then the $\forall > \exists$ reading is also true for the same situation. Thus, to tell if inverse scope is available, we should use data like (i) and (ii). In our judgement, scope reconstruction is indeed possible, and (i) and (ii) can have both readings ($\exists > \forall$, $\forall > \exists$). Still, it should be noted that getting inverse scope readings in (i) and (ii) requires the scrambled object to be contrastively interpreted. On this point, see Section 3.

- (i) **Insei-no-dareka₁-o** kaku-sensei-ga TA-tosite t₁ yatotta.
 grad-Gen-some-Acc every-teacher-Nom TA-as hired
 ‘One of the grads₁, every teacher hired t₁ as a TA.’ ($\forall \exists > \forall$, $\forall \forall > \exists$)
- (ii) **Insei-no-dareka₁-o** kaku-sensei-ga [TA-tosite t₁ yatotta]-to itta.
 grad-Gen-some-Acc every-teacher-Nom TA-as hired-C said
 ‘One of the grads₁, every teacher said that s/he hired t₁ as a TA.’ ($\forall \exists > \forall$, $\forall \forall > \exists$)

subject of the embedded finite clause is phonologically null and co-refers with the matrix subject. This change is methodologically sound, since it not only renders the pairs of CI- and LD-scrambling examples close to minimal pairs, but also eliminates some processing complexity of LD-scrambling sentences in two ways. First, making the embedded subject covert allows us to avoid a sequence of nominative markers (-ga); in general, Japanese speakers do not like such a -ga sequence as found in (4b), where the matrix and embedded nominative nouns occur adjacent. Second, using the embedded subject whose referent is the matrix one allows us to reduce the number of nominal participants to construe; this sort of complexity is also unnecessary to test whether LD-scrambling has semantic effects. In these two respects, therefore, our examples in (6b), (8b), and (10b) make it easier to concentrate on processing the interpretation of LD-scrambling.

Let us then consider the second reason to claim that the higher copy created by LD-scrambling can be present at the level of interpretation. The diagnosis we will use here is Condition C. Basically, Condition C says that the referents of a pronoun and an R-expression must be disjoint if the former c-commands the latter. As noted by Nishigauchi (2002), Mihara (2023), and Yagi (to appear), however, CI-scrambling can obviate such disjointness effects. In (11a), for example, the subject pronoun *kare*₁ ‘he’ cannot refer to the proper name *Ken*₁ buried in the object, but that is made possible by CI-scrambling, as shown in (11b).

(11) **CI-scrambling**

- a. * Kare₁-wa [Yui-ga totta Ken₁-no-syasin]-o totemo kiniitteiru.
 he-Top Yui-Nom took Ken-of-photo-Acc very like
 ‘He₁ likes [the photos of Ken₁ that Yui took] very much.’
- b. [Yui-ga totta Ken₁-no-syasin]₂-o kare₁-wa *t*₂ totemo kiniitteiru.
 Yui-Nom took Ken-of-photo-Acc he-Top very like
 ‘[The photos of Ken₁ that Yui took]₂, he₁ likes *t*₂ very much.’

What we then emphasize is that the same pattern is also observed for LD-scrambling, which is another point demonstrated by Nishigauchi (2002) and Yagi (to appear). This fact is made clear by the contrast between (12a) and (12b), the latter of which has a reading where *Ken*₁ and *kare*₁ co-refer.

(12) **LD-scrambling**

- a. * Kare₁-wa [ryosin-ga [Yui-ga totta Ken₁-no-syasin]-o kiniitteiru]-to itta.
 he-Top parent-Nom Yui-Nom took Ken-of-photo-Acc like-C said
 ‘He₁ said that his parents like [the photos of Ken₁ that Yui took].’
- b. [Yui-ga totta Ken₁-no-syasin]₂-o kare₁-wa [ryosin-ga *t*₂ kiniitteiru]-to itta.
 Yui-Nom took Ken-of-photo-Acc he-Top parent-Nom like-C said
 ‘[The photos of Ken₁ that Yui took]₂, he₁ said that his parents like *t*₂.’

Thus, the possibility of coreference in (12b) refutes the idea that LD-scrambling *obligatorily* undergoes syntactic reconstruction (i.e., higher-copy deletion). Of course, syntactic reconstruction should be kept as an option, and we need to assume that it may *optionally* apply to LD-scrambling as well as to CI-scrambling; otherwise, we could not capture the possibility of variable binding in (1B) and (2B).² Given this, we conclude that LD-

² The type of reconstruction illustrated in (1B) and (2B) is not semantic reconstruction (i.e., reconstruction through higher-type trace conversion), since it is assumed that semantic reconstruction does not feed variable binding. For this point, see Poole and Keine (2024), Yagi (to appear), and references therein. Following these researchers, we assume that syntactic and semantic reconstruction are both available, in principle.

scrambling is similar to CI-scrambling in that neither of them needs to undergo syntactic reconstruction.

3. An Alternative to the Unified Approach to Scrambling

As we have seen, CI- and LD-scrambling can both be semantically meaningful. Thus, based on this similarity, one might suggest that at least a half of Saito’s (2003) proposal is correct; that is, the two types of scrambling can be unified as one and the same movement operation. In this paper, however, we do not assume so. One piece of evidence against their unification comes from the comparison with a cross-linguistically attested type of movement, namely *topicalization*. Specifically, we will show that CI- and LD-scrambling are not uniform in that only the latter is required to obey the same application condition as topicalization does.

First of all, we note that our term “topicalization” is somewhat ambiguous. This is because we use the term to refer to a variety of operations that target the CP layer, thus including not only “topic fronting,” but also “focus fronting.” Still, while topic and focus fronting look different in nature at first glance, the two types of movement should be analyzed as making a natural class, since they are identical in one respect; that is, both of them can only apply to contrastively interpreted elements, as convincingly demonstrated by Neeleman and Vermeulen (2012). Therefore, although we refer the reader to Bianchi and Frascarelli (2010), Constant (2014), Poole (2017), Titov (2020), and Matsumoto (2025) for some relevant clarifications, we keep adopting the term “topicalization” to cover both topic fronting and focus fronting.

With this in mind, let us see how the contrast condition on topicalization is upheld. One diagnosis offered by Neeleman and Vermeulen (2012) is to use partial-answer contexts, which make topic fronting applicable. In (13), for example, A’s utterance introduces some given popstars as a new topic and asks a question about them, but B’s utterance makes an incomplete answer to that question by mentioning *the female popstars*, which is only a subset of the topic in question (i.e., *the popstars*). In short, B’s answer makes the referent of *the female popstars* contrastive with its complement set (i.e., *the male popstars*) and allows it to undergo topic fronting.

(13) A: What about **the popstars**? Who showed them around?

B: Well, **the female popstars**, Bill gave a tour. (Neeleman and Vermeulen 2012: 20)

Importantly, this type of topicalization becomes awkward in non-contrastive contexts. This is shown in (14), where the fronted noun *the female popstars* denotes the same set as the topic in question does.

(14) A: Tell me about **the female popstars**.

B: # Well, **the female popstars**, Bill gave a tour. (Neeleman and Vermeulen 2012: 20)

Thus, while the application of topic fronting is generally optional, it is the case that the movement operation can only apply to contrastively interpreted elements.³

Furthermore, the same contrast condition also restricts the application of focus fronting, as Neeleman and Vermeulen (2012) discuss in detail. One way to make this point is to use correction contexts, and we borrow discourse samples from Titov (2020), such as (15). In this exchange, B’s utterance is intended to point out there’s being a mistake in A’s utterance and provides an alternative referent that replaces the mistake.

³ Matsumoto (2025) proposes that contrastive topics always move to the CP layer and that the apparent optionality of topic fronting is reduced to which copy of them is pronounced; that is, choosing the higher copy results in overt movement, while choosing the lower copy makes it covert. What is interesting about Matsumoto’s chain reduction mechanism is that it is made restrictive enough under Richards’ (2016) *Contiguity Theory*.

- (15) A: John met Sue.
 B₁: No, John met **Mary**, not **Sue**.
 B₂: No, **Mary** John met, not **Sue**. (Titov 2020: 347)

While this correction response does not force focus fronting to apply as in (15B₁), it does make the movement operation applicable as in (15B₂), especially if it is made explicit what the fronted focus is contrasted with; here, the referents of *Mary* and *Sue* are contrasted. Thus, when no such contrastive interpretation is assigned, nothing can be fronted, even if it serves as a focus that answers a *wh*-question. This fact is shown in (16).

- (16) A: Who did John meet?
 B₁: John met **Mary**.
 B₂: # **Mary** John met. (Titov 2020: 347)

It is therefore clear that what licenses topicalization is not focus or topic interpretations but contrastive interpretations. Specifically, although contrastive elements do not obligatorily undergo focus or topic fronting, it is true that the movement operations may only apply to such elements.

Given the observations above, we now argue that LD-scrambling in Japanese must satisfy the same contrast condition as topicalization does, but CI-scrambling need not. The first half of our claim is actually nothing new, and similar ideas have been proposed by Vermeulen (2012) and Mihara (2023). However, our new point here is that, while LD-scrambling cannot be applied to non-contrastive elements, CI-scrambling is not restricted in such a way and can also be applied to non-contrastive ones. In this vein, we basically agree with Abe (2022), who reaches a similar conclusion, suggesting that LD- and CI-scrambling are not uniform in nature.

To illustrate this point, let us examine the behavior of LD- and CI-scrambling within correction contexts, which facilitate focus fronting.⁴ In (17), for example, the underlined elements are contrasted (i.e., *Bill* vs. *Peter*), and such an element can undergo LD-scrambling as in (17B₁), or may stay in situ as in (17B₂).

(17) **LD-scrambling**

- A: Zyaa, kimi-wa [Ken-ga BILL-o nagutta]-to omotteru-ndane?
 then you-Top Ken-Nom Bill-Acc hit-C think-Con(firmation)
 ‘Then, you think that Ken hit Bill, right?’

⁴ In testing our claim, we avoid using partial-answer contexts, which facilitate topic fronting, because they bring some complication into data. Specifically, once a noun is licensed as the topic of a partial answer, it must bear the topic marker *-wa*, which independently requires its bearer to occur at the front unless it serves as a focus. In (i), for example, the contrastive object in (iB₁) is not a focus (since it is not the answer part of the *wh*-question), so it cannot have an accusative case, while (iB₂) shows that such a contrastive non-focus object cannot stay in situ. For relevant discussions about this point, see Vermeulen (2012) and Matsumoto (2025).

- (i) A: Zyaa, dare-ga **aidoru-no-hitotati**-o annai-sita-no?
 then who-Nom popstar-of-people-Acc tour-did-Q
 ‘Then, who showed the popstars around?’
 B₁: Etto, ZYOSEI-aidoru{-wa/#-o} Bill-ga annai-si-masita.
 well female-popstar{-Top/-Acc} Bill-Nom tour-do-Polite.Past
 ‘Well, the female popstars, Bill showed around.’
 B₂:# Etto, Bill-ga ZYOSEI-aidoru{-wa/-o} annai-si-masita.
 well Bill-Nom female-popstar{-Top/-Acc} tour-do-Polite.Past
 ‘Well, Bill showed the female popstars around.’

- B₁: Iya, soo-zyanaku-te, PETER-o boku-wa [Ken-ga nagutta]-to omotteru-ndesu.
 no so-be.not-& Peter-Acc I-Top Ken-Nom hit-C think-Emp
 ‘No, it’s not so; Peter I think that Ken hit.’
- B₂: Iya, soo-zyanaku-te, boku-wa [Ken-ga PETER-o nagutta]-to omotteru-ndesu.
 no so-be.not-& I-Top Ken-Nom Peter-Acc hit-C think-Emp
 ‘No, it’s not so; I think that Ken hit Peter.’

The same pattern also holds for CI-scrambling. This is shown in (18), where the contrasted object in the response can be CI-scrambled as in (18B₁), or may stay in situ as in (18B₂).

(18) **CI-scrambling**

- A: Zyaa, kyoo-wa Ken-ga BILL-o nagutta-ndane?
 then today-Top Ken-Nom Bill-Acc hit-Con
 ‘Then, today Ken hit Bill, right?’
- B₁: Iya, soo-zyanaku-te, kyoo-wa PETER-o Ken-ga nagutta-ndesu.
 no so-be.not-& today-Top Peter-Acc Ken-Nom hit-Emp
 ‘No, it’s not so; today Peter Ken hit.’
- B₂: Iya, soo-zyanaku-te, kyoo-wa Ken-ga PETER-o nagutta-ndesu.
 no so-be.not-& today-Top Ken-Nom Peter-Acc hit-Emp
 ‘No, it’s not so; today Ken hit Peter.’

Of particular importance here is the fact that (17B₁) and (18B₁) are both acceptable, which shows that LD- and CI-scrambling may both apply to contrastive elements.

On the other hand, the two types of scrambling differ in that LD-scrambling cannot be applied to non-contrastive elements. In (19), for example, the embedded object *Peter* is not contrasted with anything else, and this context blocks it from undergoing LD-scrambling as in (19B₁).

(19) **LD-scrambling**

- A: Zyaa, kimi-wa [YUI-ga *Peter*-o nagutta]-to omotteru-ndane?
 then you-Top Yui-Nom Peter-Acc hit-C think-Con
 ‘Then, you think that Yui hit Peter, right?’
- B₁: #Iya, soo-zyanaku-te, *Peter*-o boku-wa [KEN-ga nagutta]-to omotteru-ndesu.
 no so-be.not-& Peter-Acc I-Top Ken-Nom hit-C think-Emp
 ‘No, it’s not so; Peter I think that Ken hit.’
- B₂: Iya, soo-zyanaku-te, boku-wa [KEN-ga *Peter*-o nagutta]-to omotteru-ndesu.
 no so-be.not-& I-Top Ken-Nom Peter-Acc hit-C think-Emp
 ‘No, it’s not so; I think that Ken hit Peter.’

However, the same sort of context does not restrict the application of CI-scrambling. This is shown in (20), and the non-contrasted object in the response can be CI-scrambled as in (20B₁).

(20) **CI-scrambling**

- A: Zyaa, kyoo-wa YUI-ga *Peter*-o nagutta-ndane?
 then today-Top Yui-Nom Peter-Acc hit-Con

B₁: Iya, soo-zyanaku-te, kyoo-wa *Peter-o* **KEN**-ga nagutta-ndesu.
no so-be.not-& today-Top Peter-Acc Ken-Nom hit-Emp
'No, it's not so; today Peter Ken hit.'

B₂: Iya, soo-zyanaku-te, kyoo-wa **KEN**-ga *Peter-o* nagutta-ndesu.
no so-be.not-& today-Top Ken-Nom Peter-Acc hit-Emp
'No, it's not so; today Ken hit Peter.'

In Shimamura and Tanaka (to appear), we have developed a specific approach to that issue, adopting a syntactic extension of *Neo-Davidsonian Event Semantics* (e.g., Borer 2005, Basilico 2008, Alexiadou 2014, Lohndal 2014, Smith and Yu 2021, Ahn 2022); for the details of our project, see also Shimamura and Tanaka (2025) and Tanaka and Shimamura (2025). In a nutshell, the verbal and nominal semantics that we propose for Japanese makes it possible to assume that, while LD-scrambling is a case of topicalization, CI-scrambling can be derived, not only by topicalization, but also by base-generation, as schematized in (21).

- Given this conclusion, we will then consider how the hybrid nature of scrambling can be further supported. To this end, our proposal offers two testable predictions. First, the semantic behavior of topicalization should

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be the same as that of LD-scrambling, because they are identified as the same type of operation that targets the CP layer. Second, given that CI-scrambling can be a case of base-generation within VoiceP, it is expected to have some configurations where it cannot exhibit reconstruction effects. These two predictions are the topics of the next two sections, and we will argue that both of them are indeed borne out.

4. Further Similarities between LD-scrambling and Topicalization

As we have shown above, LD-scrambling in Japanese can be semantically effective, as opposed to the traditional consensus. For instance, it has the ability to create a new variable binding relation as in (8), extend quantifier scope as in (10), and repair Condition C violations as in (12). Assuming this nature of LD-scrambling, our proposal predicts that topicalization in English should have the same semantic effects, since we unify it with LD-scrambling. In the following, we demonstrate that this description is basically correct.

First, topicalization can also create a new variable binding relation. This amounts to saying that topicalization obviates the effects of *weak crossover* (WCO). For instance, Guéron (1984), Lasnik and Stowell (1991), Postal (1993), and Safir (1996) note that *wh*-objects are prevented from binding pronouns within subjects as in (22a), but such a WCO effect is absent for topicalized objects. Thus, in (22b), the topicalized proper noun *John*₁ and the pronoun *his*₁ may refer to the same individual.

- (22) a. * **Who**₁ does [his₁ mother] like *t*₁?
 b. **John**₁ [his₁ mother] likes *t*₁.
 c. [His₁ mother] likes **John**₁. (Guéron 1984: 153, 163)

Still, as Ruys (2004) discusses in detail, the interpretation of *his*₁ in (22b) may be not an instance of binding, but a mere case of coreferential anaphora such as found in (22c). Accordingly, it is better to examine what happens if topicalization applies to non-referential nouns such as quantifiers. Although Lasnik and Stowell (1991) implies that quantifiers resist topicalization, Postal (1993) demonstrates that they can undergo it if they are modified by exceptive markers (e.g., *else*), relative clauses, or adjectives, as shown in (23).

- (23) a. [**Anyone {else / but Bob/ other than her}**]₁, they would have fired *t*₁.
 b. [**Anyone who was sick**]₁, they would have fired *t*₁.
 c. [**Anyone less popular**]₁, they would have fired *t*₁. (Postal 1993: 541, 542)

Given this, Pica and Snyder (1995), Marquis (1996), Eilam (2011), and Bruening (2022) observe that such a topicalized quantifier can create a new variable binding relation. In particular, Eilam (2011) convincingly argues that no WCO effect arises if the quantifier is assigned a topic interpretation and the element containing the bound pronoun is construed as a focus. This fact is shown in (24), which is cited from Marquis (1996: 323), and makes it clear that topicalization is similar to LD-scrambling in that neither is sensitive to WCO effects.⁶

⁶ Postal (1993: 542) himself claims that topicalized quantifiers cannot obviate WCO effects. According to him, for example, (i) is unacceptable on the reading where the pronoun *his*₁ is bound by *everybody else*₁.

- (i) * **Everybody else**₁, I told [his₁ wife] that I had called *t*₁. (Postal 1993: 542)

Still, Pica and Snyder (1995: 339) report that the claimed judgment for (i) is not at all clear for them. Moreover, Eilam (2011: 165) suggests that (i) can be deviant for some independent reasons. One of them is that (i) includes the quantifier *everybody*, which inherently has difficulty getting interpreted distributively. Thus, using a

- (24) a. [**Anyone who was sick**]₁, [his₁ father] would take care of *t*₁.
 b. [**Each man who was sick**]₁, [his₁ doctor] decided to stay with *t*₁ overnight.
 c. [**Each of these boys**]₁, I am sure [his₁ father] takes time to talk to *t*₁ every day.

Next, topicalization can also extend quantifier scope. Some illustrations of this capacity are offered by Bianchi and Frascarelli (2010) and Neeleman and van de Koot (2012). For example, consider the data in (25), which are cited from Bianchi and Frascarelli (2010: 69). In (25a), the embedded universal quantifier undergoes clause-internal topicalization, whereas in (25b), it crosses over the matrix existential quantifier.

- (25) a. A mechanic said [that **every one of the motorbikes**, he will fix *t* in one day]. ($\forall > \exists = 0/28$)
 b. **Every one of the motorbikes**, a mechanic said [that he will fix *t* in one day]. ($\forall > \exists = 13/28$)

There were 28 speakers who were asked to judge the available readings of these sentences. For (25a), none of the 28 speakers accepted the ' $\forall > \exists$ ' reading, but for (25b), 13 of the 28 speakers accepted it. Thus, though it looks like there is speaker variation as to the status of (25b), the difference between (25a) and (25b) is significant according to Bianchi and Frascarelli's statistical analysis. It can therefore be concluded that topicalization patterns with LD-scrambling in this respect, too, and both of them can extend quantifier scope.

Third, topicalization can also repair Condition C violations. This fact has been pointed out by Guéron (1984), Heycock (1995), and Safir (1999), among others. As shown in the following examples, when a proper name is c-commanded by a co-indexed pronoun, their referents must be disjoint, but once the proper name is embedded in a topicalized element, it is allowed to co-refer with the pronoun.

- (26) a. * He₁ has always denied [the rumour that **John**₁ stole the money].
 b. [The rumour that **John**₁ stole the money]₂, he₁ has always denied *t*₂. (Guéron 1984: 150)
- (27) a. * She₁ would never betray [**Sally**₁'s worst enemy].
 b. ? [**Sally**₁'s worst enemy]₂, she₁ would never betray *t*₂. (Heycock 1995: 553)
- (28) a. [Most articles about **Mary**₁]₂ I am sure she₁ hates *t*₂.
 b. [That **Ed**₁ was under surveillance]₂ he₁ never realized *t*₂. (Safir 1999: 553)

Thus, these examples make the final piece of evidence that our attempt to unify topicalization and LD-scrambling is on the right track; both of them can not only create a new binding relation or extend quantifier scope, but also repair Condition C violations. Given that topicalization and LD-scrambling also obey the same contrast condition for application, the data above support our claim that they are the same type of movement.

5. The Identity of CI-scrambling: Base-generation or Topicalization

Finally, we will show that our hybrid approach to scrambling also makes correct predictions on the nature of CI-scrambling. To start with, recall that CI-scrambling can be derived by base-generation or topicalization under our proposal. It is then important to note that when CI-scrambling targets a non-initial, clause-medial

different quantifier yields an acceptable sentence such as (ii), where quantifier binding is possible.

- (ii) **Every senior VP**₁, I told [his₁ secretary] that I had called *t*₁. (Eilam 2011: 166)

position, it must be a case of base-generation, since if it were topicalization, it would target the initial position (i.e., the edge of CP). Given this point, now suppose that an object noun has undergone CI-scrambling within VP; this is an instance of so-called VP-internal (VI) scrambling. In such a case, we predict that the scrambled object can only be interpreted in its overt position, and not in any lower ones, because the scrambling in question involves no movement, hence no reconstruction. This prediction is indeed borne out.

To illustrate, let us examine the behavior of VI-scrambling in the ditransitive construction. A relevant observation is made by Nemoto (1993: 91). What she shows is that the dative noun (Dat) can bind the reciprocal *otagai* within the accusative noun (Acc) in the Dat-Acc order, as in (29a), but VI-scrambling of Acc over Dat makes the binding impossible, as in (29b).

(29) **VI-scrambling**

- a. Boku-wa tyokusetu [Ken-to-Yui]₁-ni **otagai₁-no-sukina-taipu-o** tutaeta.
 I-Top directly Ken-&-Yui-Dat e.o-of-favorite-type-Acc told
 ‘I told [Ken and Yui]₁ each other₁’s favorite type directly.’
- b. * Boku-wa tyokusetu **otagai₁-no-sukina-taipu-o** [Ken-to-Yui]₁-ni tutaeta.
 I-Top directly e.o-of-favorite-type-Acc Ken-&-Yui-Dat told
 ‘I told each other₁’s favorite type to [Ken and Yui]₁ directly.’

The status of (29b) is mysterious for an analysis that derives the Acc-Dat order by movement, since it should allow Acc to reconstruct. For our analysis, however, it is not mysterious at all, because VI-scrambling is a result of base-generation and involves no movement (cf. Miyagawa 1997).

Now, consider what happens if scrambling of Acc targets the clause-initial position. We predict that it allows Acc to reconstruct, since it can be a case of movement, i.e., topicalization. Though Nemoto (1993: 93) denies the availability of such a reconstruction reading, we find it totally possible if the scrambled Acc is interpreted contrastively. This is shown by the following discourse, where (30B) serves as a corrective answer to (30A) and thus facilitates focus fronting of contrastive elements.

(30) **CI-scrambling**

- A: Etto, kimi₂-wa [Ken-to-Yui]₁-ni **ZIBUN₂-no-sukina-taipu-o** tutaeta-no?
 well you-Top Ken-&-Yui-Dat self-of-favorite-type-Acc told-Q
 ‘Well, did you₂ tell [Ken and Yui]₁ YOUR₂ favorite type?’
- B: Iya, **OTAGAI₁-no-sukina-taipu-o** boku₂-wa [Ken-to-Yui]₁-ni tutaeta-ndayo.
 no e.o-of-favorite-type-Acc I-Top Ken-&-Yui-Dat told-Emp
 ‘No, EACH OTHER₁’s favorite type, I₂ told [Ken and Yui]₁.’

Our base-generation analysis of VI-scrambling is further supported by other kinds of diagnoses. In (31a) and (31b), we check whether variable binding is available. (31a) shows that the Dat-Acc order enables the dative quantifier to bind the pronoun *soko*₁ in the accusative noun, but (31b) shows that the “scrambled” Acc-Dat order cannot convey such a reading, as pointed out by Yagi (to appear) (cf. Hoji 1985).

(31) **VI-scrambling**

- a. Boku-wa meeru-de kaku-tenpo₁-ni **soko₁-no-warui-bubun-o** tutaeta.
 I-Top email-by every-shop-Dat there-of-bad-part-Acc told
 ‘I told every shop₁ its₁ weaknesses by email.’

- b. * Boku-wa meeru-de **soko₁-no-warui-bubun-o** kaku-tenpo₁-ni tutaeta.
 I-Top email-by there-of-bad-part-Acc every-shop-Dat told
 ‘I told its₁ weaknesses to every shop₁ by email.’

On the other hand, scrambling of Acc over the subject makes variable binding available again, if Acc is contrastively construed, as illustrated in (32B).

(32) **CI-scrambling**

- A: Etto, kimi-wa kaku-tenpo₁-ni **soko₁-no-YOI-bubun-o** tutaeta-no?
 well you-Top every-shop-Dat there-of-good-part-Acc told-Q
 ‘Well, did you tell every shop₁ its₁ STRENGTHS?’
 B: Iya, **soko₁-no-WARUI-bubun-o** boku-wa kaku-tenpo₁-ni tutaeta-ndayo.
 no there-of-bad-part-Acc I-Top every-shop-Dat told-Emp
 ‘No, its₁ WEAKNESSES, I told every shop₁.’

The same point is also made by scope interaction. As the comparison of (33a) and (33b) shows, the Acc-Dat order makes it quite difficult to get a reading where Acc takes scope below Dat (cf. Hoji 1985, Yagi to appear).

(33) **VI-scrambling**

- a. Boku-wa sokutatu-de kaku-seito-ni **Sundai-mosi-no-hitotu-o** okutta.
 I-Top express-by every-pupil-Dat Sundai-exam-of-one-Acc sent
 ‘I sent every pupil one of the Sundai mock exams by express.’ ($\forall > \exists$)
 b. Boku-wa sokutatu-de **Sundai-mosi-no-hitotu-o** kaku-seito-ni okutta.
 I-Top express-by Sundai-exam-of-one-Acc every-pupil-Dat sent
 ‘I sent every pupil one of the Sundai mock exams by express.’ ($?*\forall > \exists$)

On the other hand, (34B) is a case of scrambling of Acc to the clause-initial position. We judge it as rendering an inverse scope reading much easier to obtain, as long as the scrambled Acc bears a contrastive interpretation.

(34) **CI-scrambling**

- A: Etto, kimi-wa kaku-seito-ni **KAWAI-mosi-no-hitotu-o** okutta-no?
 well you-Top every-pupil-Dat Kawai-exam-of-one-Acc sent-Q
 ‘Well, did you send every pupil one of the KAWAI mock exams?’
 B: Iya, **SUNDAI-mosi-no-hitotu-o** boku-wa kaku-seito-ni okutta-ndayo.
 no Sundai-exam-of-one-Acc I-Top every-pupil-Dat sent-Emp
 ‘No, one of the SUNDAI mock exams, I sent every pupil.’ ($\forall > \exists$)

In this way, CI-scrambling exhibits dual nature; its interpretation varies according to the position it targets. When it targets a clause-medial position as in the case of VI-scrambling, it cannot allow the scrambled object to reconstruct and forces it to be interpreted in its overt position. On the other hand, when CI-scrambling targets the initial position with the scrambled object construed contrastively, it allows the object to reconstruct and to be interpreted in lower positions than its overt one. This dual nature of CI-scrambling perfectly fits our proposal, since it assumes that CI-scrambling can be derived by base-generation or topicalization. That is, under our theory, CI-scrambling to a medial position must be a case of base-generation, so it can undergo no reconstruction,

while CI-scrambling to the initial position can be a case of topicalization, so it makes reconstruction available. These two different derivations of CI-scrambling are schematized below.

- (35) a. **CI-scrambling over the dative must be base-generation in VP** (no reconstruction)
 [CP ... [... [VoiceP Subj [VP **Acc** Dat V] ...] C]
 ↑ *direct base-generation*
- b. **CI-scrambling over the subject can be topicalization into CP** (^{OK} reconstruction)
 [CP **Acc**₁ [... [VoiceP Subj [VP Dat **t**₁ V] ...] C]
 ↑ *topic/focus movement* ———

6. Conclusion

In summary, this paper made two new points on the nature of scrambling in Japanese. First, we showed that not only CI-, but also LD-scrambling can create new binding/scope relations, so the latter does not obligatorily undergo radical reconstruction, either (cf. Saito 1992, Tada 1993). Second, we argued that, though CI- and LD-scrambling can both have semantic effects, they are actually not uniform in nature and must be derived by different sets of syntactic processes (cf. Tada 1990, Kitahara 2002, Saito 2003). Specifically, we revealed that, for LD-scrambling to apply, it must satisfy the same contrast condition as English topicalization does, but CI-scrambling need not. To explain this difference, we built on our proposal in Shimamura and Tanaka (to appear), under which LD-scrambling is always a case of topicalization, but CI-scrambling can be derived by topicalization or base-generation. We lent additional support to this dual identity of CI-scrambling by showing that VI-scrambling and CI-scrambling over the subject behave differently (cf. Nemoto 1993). We hope that the new empirical picture established in this paper will stimulate further investigations into scrambling and reconstruction in Japanese (e.g., Yagi to appear) or other languages such as Hindi-Urdu (e.g., Poole and Keine 2024).

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